

Topics

1. What is Variable
2. How to declare variable
3. Naming convention of a variable
4. Types of Variable:

4.1 - Primitive Type

- boolean Type
- boolean

4.2 - Numeric Type

- Integral Type
 - byte
 - char
 - short
 - int
 - long

- Floating Type

- float

- double

- Reference Type

- classes

- Interfaces

- Array

- String

- Enum

- Kind of variables:

- local

- Instance

- Static/class variable

- Method parameters

- constructor Parameters

- Types of Conversion:

- Widening / Automatic Conversion

- Narrowing / Downcasting / Explicit Conversion

- Promotion during expression

- Explicit Casting during expression

- Wrapper classes

- Boxing / Autoboxing and Unboxing

- final variables

- static final variable (similar to constant in C) only 1 copy and immutable.

1. Variable is a container which holds a value

2. Syntax: `Datatype VariableName = value;`

Java is a static typed as well as Strong Typed lang.
 means $\rightarrow$ Every variable has a datatype which tell what
 kind of value^{data} it will store

(datatype val)
 $\rightarrow$ `int val`

Strong type lang $\rightarrow$ every datatype has a range of values defined it can't store value beyond the range, that has some restriction.

`int var = 300`
 $\downarrow$
 range [lb, ub]

3. Variable naming convention:

1. Variable name is case-sensitive

2. Variable name can be any legal identifier means can contain Unicode letters
 $\rightarrow$ all language characters, digit

3. Variable name can't be start with special symbol except \$, -, it can start with letters also.

4. keywords cannot be used as variable name

5. Variable name should be all small if it contain only 1 word else camel case should be followed

Types of Variables

Java has 8 primitive datatypes like [int, char, long, float, double, boolean, short, byte.

- char : 2 bytes (16 bits)

• character repⁿ of ASCII value

• Range : 0 to 65535 i.e. '\u0000' to '\uffff'

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- byte : 1 byte (8 bits)
Signed 2's complement
Range : -128 to 127 $\xrightarrow{\text{-ve}} \xrightarrow{\text{+ve}}$ total 255 values
default value is 0

- Short : 2 byte (16 bits)
Signed 2's complement
Range : -32768 to 32767 (65535 values)
default value is 0.

- Long : 8 byte (64 bits)
Signed 2's complement
Range -2^{63} to $2^{63}-1$
default value is '0L'

- int : 4 bytes (32 bits)
Signed 2's complement
Range -2^{31} to $2^{31}-1$
default value is 0.

Floating / fractional value

- float $\rightarrow$ 32 bit IEEE 754 values

Ex: float var = 63.20f;

Solve: float var1 = 0.3f

float var2 = 0.1f

cout << (var1 - var2); // we expect 0.2f as Ans but we get 0.20002f

that's why no big company who is working with currency doesn't use float & double for decimal calcn, they use BigInteger

- double $\rightarrow$ 64 bit IEEE 754 value
default value

- boolean $\rightarrow$ 1 bit
 $\rightarrow$ True / False

Types of conversions

1. Widening / Automatic conversion: whenever you go from lower to higher automatic convs happens.

byte (1 byte) $\downarrow$
 short (2 byte) $\downarrow$
 int (4 byte) $\downarrow$
 long (8 byte) $\downarrow$
 float (4 byte) $\downarrow$
 double (8 byte) $\downarrow$

automatic convsⁿ

int var = 100

long varlong = var
✓ correct

2. Downcasting / Explicit conversion: when comes from larger DT to smaller DT then automatic convs not possible, we have to manually tell the compiler for conversion, in that way we lost some precision of data.

Ex int var1 = 128;

byte byteVar = (byte)var1 // -128 (Range is -128 to 127)
 after reach 127 it again starts to lower range value

3. Promotion during Exprⁿ / explicit type convⁿ.

Ex int a = 127

double b = 3.0;

int res = a + b; // gives error

double res = a + b ✓ work fine because final result is ~~after~~ behaves like larger datatype used in expression. which means result got promoted to double.

Types of variables

Ex:

```
class Employee {
```

```
    int memberVariable; // member variable
```

```
    static int staticVariable = 100; // static variable or class variable
```

```
    Employee() { }
```

```
    Employee(int a) { // Parameterized constructor
```

```
        this.memberVariable = a;
```

```
    }
```

```
    public voidint dummyMethod(int a, int b) { // local variable
```

```
        return a + b;
```

```
    }
```

```
}
```

```
public class Main {
```

```
    PSUM() {
```

```
        Employee e1 = new Employee();
```

```
        Employee e2 = new Employee();
```

```
    }
```

```
}
```

